

ation. Variation happens readily in such markets. In contrast, selection is difficult because it is challenging to figure out which experience should be generalized from the extensive situation-specific knowledge that occurs. Which of the many experiences should be incorporated into the ongoing routines for the capabilities and which should be forgotten? The temptation is to generalize too quickly, and so to churn capabilities too often on the basis of idiosyncratic events (Gersick, 1994; Sastry, 1999).

Finally, the order of implementation of dynamic capabilities is consequential. That is, dynamic capabilities are often combinations of simpler capabilities and related routines, some of which may be foundational to others and so must be learned first. Brown and Eisenhardt (1997) termed this property 'sequenced steps.' In their study of multiple product development processes in six firms in the computer industry, they observed that multiple product development required the combination of three simpler dynamic capabilities: single product development, probing the future and linking routines from one product development project to the next. Managers who built an effective dynamic capability to develop multiple products did so according to 'sequence steps' that had to be executed in the proper order. Single product development skills needed to come first to provide the platform for future products, then skills related to probing the future for new product opportunities, and finally time-pacing skills to create a product development rhythm connecting current products to future ones. Similarly, in his study of Hyundai, Kim (1998) found an appropriate sequencing of the learning of capabilities from the simpler and more predictable capabilities around manufacturing process creation to the more improvisational design ones in the development of design routines. Thus, effective implementation requires knowing both the ingredients (i.e., key commonalities of capabilities) and the recipe (i.e., order of implementation).

DISCUSSION

The purpose of this paper is to explore dynamic capabilities and more generally, RBV. In addressing this agenda, we focused on the nature of dynamic capabilities, the impact of market dyna-

mism, and their evolution. Our observations link to several research areas.

Our work suggests reframing the concept of dynamic capabilities. Dynamic capabilities are not tautological, vague, and endlessly recursive as some have suggested (e.g., Priem and Butler, 2000; Williamson, 1999). Rather, they consist of many well-known processes such as alliancing, product development, and strategic decision making that have been studied extensively in their own right, apart from RBV. Their value for competitive advantage lies in their ability to alter the resource base: create, integrate, recombine, and release resources.

Dynamic capabilities also exhibit commonalities across firms that are associated with superior effectiveness. So while the specifics of any given dynamic capability may be idiosyncratic to a firm (e.g., exact composition of a cross-functional product development team) and path dependent in its emergence, 'best practice' exists for particular dynamic capabilities across firms. These commonalities imply that dynamic capabilities are equifinal such that firms can develop these capabilities from many starting points and along different paths. They are also more homogeneous, fungible, and substitutable than is usually assumed. Overall, these observations suggest a modified conception of dynamic capabilities.

Our work also suggests an expanded view of routines (Cyert and March, 1963; Nelson and Winter, 1982; Winter and Szulanski, 1999). We argue that, in moderately dynamic markets, routines in the form of dynamic capabilities are embedded in cumulative, existing knowledge. They involve analysis using existing knowledge and rules of thumb, followed by implementation. When this existing knowledge is codified, the resulting routines are often detailed and specific with predictable outcomes (Helfat, 1997; Nelson and Winter, 1982). Therefore, in moderately dynamic markets, dynamic capabilities exhibit the properties suggested in the traditional research where effective routines are efficient and robust processes (Cyert and March, 1963; Nelson and Winter, 1982).

In contrast, in high-velocity markets, dynamic capabilities rely extensively on new knowledge created for specific situations. Routines are purposefully simple to allow for emergent adaptation, although not completely unstructured. Since new knowledge must be rapidly gained in each new